# SG No. 179 $D_6^3$ $P6_522$ [ hexagonal ]

* generator : $\{3^+_{001}|00\frac{2}{3}\}$, $\{2_{001}|00\frac{1}{2}\}$, $\{2_{110}|00\frac{2}{3}\}$
* symmetry operation $+[0, 0, 0]$

Table 1: Symmetry operations for 3d polar vector.

| No. | tag | matrix (polar) | det |
|---|---|---|---|
| 1 | $\{1\|0\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | 1 |
| 2 | $\{3^+_{001}\|00\frac{2}{3}\}$ | $\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{2}{3} \end{bmatrix}$ | 1 |
| 3 | $\{3^-_{001}\|00\frac{1}{3}\}$ | $\begin{bmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{3} \end{bmatrix}$ | 1 |
| 4 | $\{2_{001}\|00\frac{1}{2}\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$ | 1 |
| 5 | $\{6^-_{001}\|00\frac{1}{6}\}$ | $\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{6} \end{bmatrix}$ | 1 |
| 6 | $\{6^+_{001}\|00\frac{5}{6}\}$ | $\begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & \frac{5}{6} \end{bmatrix}$ | 1 |
| 7 | $\{2_{110}\|00\frac{2}{3}\}$ | $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{2}{3} \end{bmatrix}$ | 1 |
| 8 | $\{2_{100}\|0\}$ | $\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | 1 |
| 9 | $\{2_{010}\|00\frac{1}{3}\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{3} \end{bmatrix}$ | 1 |
| 10 | $\{2_{1-10}\|00\frac{1}{6}\}$ | $\begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{6} \end{bmatrix}$ | 1 |
| 11 | $\{2_{120}\|00\frac{1}{2}\}$ | $\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$ | 1 |
| 12 | $\{2_{210}\|00\frac{5}{6}\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & \frac{5}{6} \end{bmatrix}$ | 1 |